# Byung-Kwan Lee

Ph.D., School of Electrical Engineering at KAIST

**Research Scientist** at NVIDIA

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## Research Interest

Building **efficient, high-performing Vision-Language Models (VLMs)**, with focus on:

- Distillation
- Pruning
- RL
- Agent
- Harness

## Work Experience

### NVIDIA • *Research Scientist* — *Oct. 2025 - Current*

**LEAD** **Mid-Harness: Scaling Actions Between Model and Harness** [COMPLETED] *Action-level test-time scaling for terminal agents that samples and verifies candidate commands at the model–harness boundary before execution, lifting trajectory success without changing the generator or harness.*

Under Review · U.S. Provisional Patent Filed · Internal Tech Transfer

**LEAD** **ZPPO: Zone of Proximal Policy Optimization** [COMPLETED] *RL post-training that keeps the teacher inside the prompt rather than the policy gradient, using BCQ/NCQ question reformulations and a prompt replay buffer to lift small students on hard questions without policy drift.*

Under Review · U.S. Provisional Patent Filed

**LEAD** **AXPO: Agent eXplorative Policy Optimization** [COMPLETED] *Agentic RL training that recovers tool usage through tool-call resampling, improving multimodal reasoning performance against larger baselines.*

NeurIPS 2026 · U.S. Provisional Patent Filed · Internal Tech Transfer

**LEAD** **Masking Teacher and Reinforcing Student** [COMPLETED] *Mask-progressive RL distillation that gradually unmasks teacher weights and uses offline RL with accuracy & distillation rewards.*

CVPR 2026 · U.S. Non-Provisional Patent Filed

**LEAD** **Unified RL & Imitation Learning for VLMs** [COMPLETED] *Combines RL with adversarial imitation, using an LLM-based discriminator and multi-teacher guidance to build lightweight yet powerful VLMs.*

NeurIPS 2025 · U.S. Patent Upgraded to Non-Provisional

**LEAD** **GenRecal** [COMPLETED] *Cross-architecture VLM distillation via a Recalibrator that aligns heterogeneous token representations regardless of vocabulary, token splits, or index ordering.*

ECCV 2026 · U.S. Patent Upgraded to Non-Provisional · Internal Tech Transfer

**LEAD** **VLsI: Verbalized Layers-to-Interactions** [COMPLETED] *Hardened the verbalizer-based layer-wise distillation into a production-ready recipe and led its internal tech transfer into NVIDIA's VLM development pipeline.*

CVPR 2025 · U.S. Patent Upgraded to Non-Provisional · Internal Tech Transfer

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### NVIDIA • *Research Intern* — *Oct. 2024 - Oct. 2025*

**LEAD** **Unified RL & Imitation Learning for VLMs** [INITIATED] *First formulation of the RL + adversarial imitation training pipeline; initial experiments and team setup.*

U.S. Provisional Patent Filed

**LEAD** **GenRecal** [INITIATED] *Initial design of the Recalibrator framework for cross-tokenizer VLM distillation; first proof-of-concept and internal demo.*

U.S. Provisional Patent Filed

**LEAD** **VLsI: Verbalized Layers-to-Interactions** [INITIATED] *Layer-wise distillation using intermediate verbalizers, enabling small VLMs (2B/7B) to align with large VLMs' reasoning progression and outperform GPT-4V.*

CVPR 2025 · U.S. Provisional Patent Filed

## Education

**KAIST** *Mar. 2020 - Aug. 2025*
*Ph.D., School of Electrical Engineering*
◦ GPA: 3.77/4.3

◦ Dissertation: *Building High-performing, Efficient-size Vision Language Models: Merge, Modify, and Distill* [Link]

**KAIST** *Mar. 2018 - Feb. 2020*
*M.S., The Cho Chun Sik Graduate School of Green Transportation*
◦ GPA: 3.72/4.3

◦ Thesis: *Training Encoder-Attention through Fully-Connected CRFs for Efficient End-to-End Lane Detection Model* [Link]

**Hanyang University** *Mar. 2014 - Feb. 2018*
*B.S., Mathematics and Electronic Engineering*
◦ GPA: 3.86/4.5

◦ Thesis: *Learning Rank of Evaluated Songs for Classifying Unknown Music by Convolutional Neural Network* [Link]

## Publications

| | |
|---|---|
| **Overall** | **21** Accepts · **4** Pending · **5** Tech Reports |
| **CV** | **5** CVPR · **2** ICCV · **3** ECCV · **1** ICIP |
| **ML** | **6** NeurIPS · **1** ICLR |
| **NLP** | **1** ACL · **1** EMNLP |
| **Journal** | **1** Pattern Recognition |
| **In Progress** | **16** On-going Projects |




■ **67%** (20/30) First & Lead Author
■ **33%** (10/30) Co-Author

1. **Mid-Harness: Scaling Actions Between Model and Harness for Terminal Agents**
   Minki Kang, Ryo Hachiuma, Shaokun Zhang, Subhashree Radhakrishnan, Yonggan Fu, Jindong Jiang, Mingjie Liu, Ehsan Hosseini-Asl, Yi Dong, Yu-Chiang Frank Wang, Byung-Kwan Lee

   Under Review [Paper][Project]

   U.S. Patent Application Filed (Provisional), NVIDIA Research

2. **Zone of Proximal Policy Optimization: Teacher in Prompts, Not Gradients**
   Byung-Kwan Lee, Ximing Lu, Shizhe Diao, Minki Kang, Saurav Muralidharan, Karan Sapra, Andrew Tao, Yu-Chiang Frank Wang, Ryo Hachiuma

   Under Review [Paper][Project]

   U.S. Patent Application Filed (Provisional), NVIDIA Research

3. **When Do We Need On-Policy Distillation? Distilling on Offline Student Rollouts Is Often Better**
   Siyan Zhao*, Yonggan Fu*, Jindong Jiang, Shih-Yang Liu, Song Bian, Byung-Kwan Lee, Sharath Turuvekere Sreenivas, Wenliang Dai, Hanrong Ye, Aditya Grover, Pavlo Molchanov

   Under Review [Paper]

4. **When to Switch: Reliable Action-Chunk Extension for Vision-Language-Action Models**
   Seonghoon Yu, Dongwon Kim, HyungRok Jung, Yoonjae Baek, Byung-Kwan Lee, Suha Kwak, Jeany Son

   Under Review [Paper][Code]

5. **Agent eXplorative Policy Optimization for Multimodal Agentic Reasoning**
   Minki Kang, Shizhe Diao, Ryo Hachiuma, Sung Ju Hwang, Pavlo Molchanov, Yu-Chiang Frank Wang, Byung-Kwan Lee

   Neural Information Processing Systems (NeurIPS), 2026 [Paper][Project]

   U.S. Patent Application Filed (Provisional), NVIDIA Research

6. **SpatialClaw: Rethinking Action Interface for Agentic Spatial Reasoning**
   Seokju Cho, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, Min-Hung Chen

Neural Information Processing Systems (NeurIPS), 2026 [Paper][Project][Code]

U.S. Patent Application Filed (Provisional), NVIDIA Research

7. **Hide to See: Reasoning-prefix Masking for Visual-anchored Thinking in VLM Distillation**
Seonghoon Yu, Dongjun Nam, Byung-Kwan Lee†, Jeany Son†

Neural Information Processing Systems (NeurIPS), 2026 [Paper][Project][Code]

8. **GenRecal: Generation after Recalibration from Large to Small Vision Language Models**
Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, and Yueh-Hua Wu

European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

U.S. Patent Application Filed (Non-Provisional), NVIDIA Research

9. **Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding**
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, Chanyoung Park

European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

10. **Masking Teacher and Reinforcing Student for Distilling Vision-Language Models**
Byung-Kwan Lee, Yu-Chiang Frank Wang, Ryo Hachiuma

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [Paper]

U.S. Patent Application Filed (Non-Provisional), NVIDIA Research

11. **Recursive Think-Answer Process for LLMs and VLMs**
Byung-Kwan Lee*, Youngchae Chee*, Yong Man Ro

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026 [Paper][Project]

12. **RefineBench: Evaluating Refinement Capability in Language Models**
Young-Jun Lee*, Seungone Kim*, Byung-Kwan Lee, Minkyeong Moon, Yechan Hwang, Jong Myoung Kim, Graham Neubig, Sean Welleck, Ho-Jin Choi

International Conference on Learning Representations (ICLR), 2026 [Paper][Project]

Best Runner-Up Award, Workshop for Multi-Turn Interactions in Large Language Models, NeurIPS 2025 (Oral, Top 1%) [Link]

13. **Enhancing Conversational Agents with Skill-of-Mind-Infused Large Language Model**
Young-Jun Lee, Byung-Kwan Lee, Dokyong Lee, Junyoung Youn, Kyeong-Jin Oh, Yechan Hwang, Ho-Jin Choi

Technical Report [Paper][Code][Huggingface Model]

14. **Unified Reinforcement and Imitation Learning for Vision-Language Models**
Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, Yueh-Hua Wu

Neural Information Processing Systems (NeurIPS), 2025 [Paper][Project]

U.S. Patent Application Filed (Non-Provisional), NVIDIA Research

15. **MultiVerse: A Multi-Turn Conversation Benchmark for Evaluating Large Vision and Language Models**
Young-Jun Lee, Byung-Kwan Lee, Jianshu Zhang, Yechan Hwang, Byungsoo Ko, Han-Gyu Kim, Dongyu Yao, Xuankun Rong, Eojin Joo, Seung-Ho Han, Bowon Ko, Ho-Jin Choi

IEEE/CVF International Conference on Computer Vision (ICCV), 2025 [Paper][Project]

Workshop for Knowledge-Intensive Multimodal Reasoning, ICCV 2025 [Link]

16. **Are Vision-Language Models Truly Understanding Multi-vision Sensor?**
Sangyun Chung*, Youngjoon Yu*, Youngchae Chee, Se Yeon Kim, Byung-Kwan Lee, Yong Man Ro

Technical Report [Paper][Code]

17. **VLsI: Verbalized Layers-to-Interactions from Large to Small Vision Language Models**
Byung-Kwan Lee, Ryo Hachiuma, Yu-Chiang Frank Wang, Yong Man Ro†, Yueh-Hua Wu†

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 [Paper][Project]

U.S. Patent Application Filed (Non-Provisional), NVIDIA Research

18. **Phantom of Latent for Large Language and Vision Models**
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro

Technical Report [Paper][Code][Huggingface Model]

19. **SPARK: Multi-Vision Sensor Perception and Reasoning Benchmark for Large-scale Vision-Language Models**
Youngjoon Yu*, Sangyun Chung*, Byung-Kwan Lee, Yong Man Ro

Technical Report [Paper][Code][Huggingface Dataset]

20. **TroL: Traversal of Layers for Large Language and Vision Models**
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro

Empirical Methods in Natural Language Processing (EMNLP), 2024 [Paper][Code][Huggingface Model]

21. **Meteor: Mamba-based Traversal of Rationale for Large Language and Vision Models**
Byung-Kwan Lee, Chae Won Kim, Beomchan Park, Yong Man Ro

Neural Information Processing Systems (NeurIPS), 2024 [Paper][Code][Huggingface Model]

31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering

22. **MoAI: Mixture of All Intelligence for Large Language and Vision Models**
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro

European Conference on Computer Vision (ECCV), 2024, [Paper][Code][Huggingface Model]

23. **CoLLaVO: Crayon Large Language and Vision mOdel**
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro

Findings of the Association for Computational Linguistics (ACL), 2024, [Paper][Code][Huggingface Model]

2024 KCC XAI Workshop, Best Paper Awards

24. **Causal Unsupervised Semantic Segmentation**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro

Journal of Pattern Recognition [Paper][Code]

25. **Mitigating Adversarial Vulnerability through Causal Parameter Estimation by Adversarial Double Machine Learning**
Byung-Kwan Lee*, Junho Kim*, and Yong Man Ro

IEEE/CVF International Conference on Computer Vision (ICCV), 2023 [Paper][Code]

26. **Mitigating Dataset Bias in Image Captioning through CLIP Confounder-free Captioning Network**
YeonJu Kim, Junho Kim, Byung-Kwan Lee, Sebin Shin, and Yong Man Ro

IEEE International Conference on Image Processing (ICIP), 2023 [Paper][Code]

27. **Demystifying Causal Features on Adversarial Examples and Causal Inoculation for Robust Network by Adversarial Instrumental Variable Regression**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [Paper][Code]

28. **Masking Adversarial Damage: Finding Adversarial Saliency for Robust and Sparse Network**
Byung-Kwan Lee*, Junho Kim*, and Yong Man Ro

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022 [Paper][Code]

29. **Distilling Robust and Non-Robust Features in Adversarial Examples by Information Bottleneck**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro

Neural Information Processing Systems (NeurIPS), 2021 [Paper][Code]

30. **Towards Adversarial Robustness of Bayesian Neural Network through Hierarchical Variational Inference**
Byung-Kwan Lee*, Youngjun Yu, and Yong Man Ro

Technical Report [Paper][Code]

## Reviewer Experience

| | |
|---|---|
| IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) | *Journal* |
| IEEE Transactions on Neural Networks and Learning Systems (TNNLS) | *Journal* |
| IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) | *Journal* |
| IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) | *Conference* |
| IEEE/CVF International Conference on Computer Vision (ICCV) | *Conference* |
| European Conference on Computer Vision (ECCV) | *Conference* |
| International Conference on Learning Representations (ICLR) | *Conference* |
| Neural Information Processing Systems (NeurIPS) | *Conference* |
| International Conference on Machine Learning (ICML) | *Conference* |
| AAAI Conference on Artificial Intelligence (AAAI) | *Conference* |
| Association for Computational Linguistic (ACL) | *Conference* |
| Empirical Methods in Natural Language Processing (EMNLP) | *Conference* |

## Invited Talks and Awards

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|---|---|
| Invited Talk in Chung-Ang University – Metaverse | *2026* |
| Invited Talk in Chung-Ang University | *2026* |
| Invited Talk in Kongju National University | *2026* |
| Invited Talk in Kyonggi University | *2026* |
| Invited Talk in Kookmin University | *2026* |
| Top Reviewer Award, Neural Information Processing Systems (NeurIPS) | *2026* |
| Outstanding Reviewer Award, European Conference on Computer Vision (ECCV) | *2026* |
| Silver Reviewer Award, International Conference on Machine Learning (ICML) | *2026* |
| Best Runner-Up Award (Oral, Top 1%), Multi-Turn Interactions in LLMs Workshop @ NeurIPS | *2025* |
| Invited Talk in Kongju National University | *2025* |
| Invited Talk in Hanyang University ERICA | *2025* |
| Invited Talk in Kookmin University | *2025* |
| Invited Talk in Dongseo University | *2025* |
| 31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering | *2025* |
| KCC XAI Workshop, Best Paper Awards | *2024* |
| Invited Talk in Korea Institute of Science and Technology Information (KISTI) | *2024* |
| Invited Talk in NAVER HyperCLOVA X | *2024* |
| Invited Talk in Kakao Brain | *2024* |
| 1st Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI) | *2023* |
| GIRE Research Mentor, Seoul International School – "MUsE", Journal of Student Research [Publication] | *2023* |
| KAIST SW IT Academy, AI Project Director of 1st Award Topic | *2023* |
| Invitation to Presenter in Korean Conference on Computer Vision (KCCV) | *2023* |
| KAIST SW IT Academy, Lecturer of Data Structure & Algorithm, Deep Learning, Computer Vision | *2023* |
| Invitation to Presenter in Korean Conference on Computer Vision (KCCV) | *2022* |
| 2nd Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI) | *2022* |
| KAIST Fellowship | *Mar. 2020 - Feb. 2024* |
| National Government Fellowship | *Mar. 2018 - Feb. 2020* |
| 1st President's Award for International Autonomous Vehicle Competition, Team Leader [YouTube][Talk] | *2018* |